Q/R/M/C: a four-stage decomposition of semantic memory navigation

Q — Query formation

Build a semantic query from intent and state

$Q^\star$: query has ≥ 1 candidate

R — Retrieval

Memory returns a candidate satisfying the predicate

$R^\star$: candidate satisfies predicate

M — Materialization

Planner locks onto a concrete reachable target cell

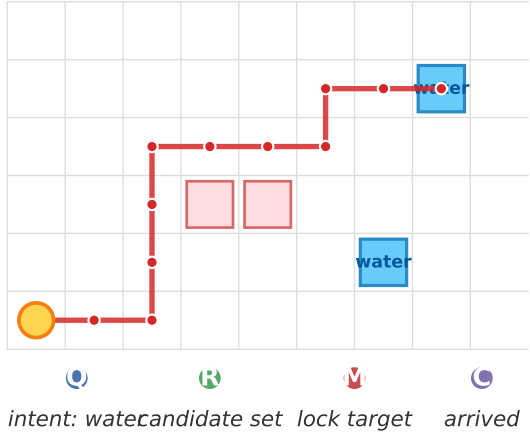
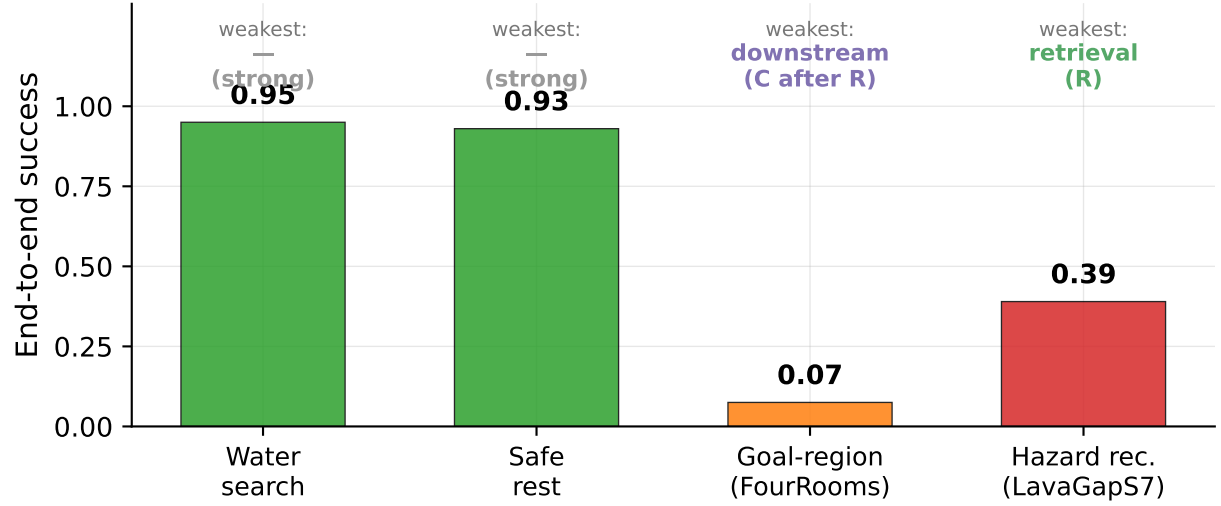
$M^\star$: target lock acquired

C — Completion

Controller executes and reaches the materialized target

$C^\star$: agent reaches target

(a) Example episode: stages anchor at observable events



(b) Same scalar success hides distinct stage failures